Regression Autopsy

Eight Ways Your Model Is Lying to You

Lecture Notes

Statistical and Machine Learning Lab

Spring 2026

These notes accompany the interactive Jupyter notebook series. All theorems stated here are validated by Monte Carlo simulation in the corresponding notebooks.

Contents

1 Introduction **3**
1.1 The NovaMart Scenario . . . 3

2 The Data Generating Process **3**
2.1 Pathologies Embedded in the DGP . . . 4

3 Why Your Coefficient Is Wrong: Omitted Variable Bias **4**
3.1 Setup . . . 4
3.2 When OVB Cannot Be Signed . . . 6
3.3 The Proxy Variable Approach . . . 6

4 Why Your Uncertainty Is Wrong: Heteroscedasticity **6**
4.1 The Classical Variance Formula . . . 6
4.2 Consequences for Inference . . . 7
4.3 The Breusch–Pagan Test . . . 7
4.4 Robust Standard Errors . . . 8

5 Why Your Significance Is Wrong: Effect Size and Power **8**
5.1 The Anatomy of a t-Statistic . . . 8
5.2 Statistical Power . . . 9
5.3 The Multiple Testing Problem . . . 9

6 Why Your Model Is Wrong: Specification Error **10**
6.1 The Problem of Functional Form . . . 10
6.2 The Extrapolation Danger . . . 10

7 Why Your R^2 Is Lying: Overfitting **11**
7.1 The Bias–Variance Decomposition . . . 11
7.2 Overfitting in Practice . . . 12
7.3 Cross-Validation . . . 12

8 Why Your Causal Claim Is Wrong: Endogeneity and Bad Controls **13**
8.1 Endogeneity . . . 13
8.2 Collider Bias from Bad Controls . . . 14
8.3 Instrumental Variables . . . 14

9 The Real World: When All Pathologies Strike at Once **15**
9.1 The Lalonde Dataset . . . 16
9.2 Sensitivity Analysis: Cinelli–Hazlett Framework . . . 16

10 The Redemption: Regression Discontinuity Design **16**
10.1 When Design Earns Trust . . . 16
10.2 Example: Scholarship Cutoff . . . 17
10.3 Bandwidth Choice . . . 17

11 What "Controlling For" Actually Means: The Frisch–Waugh–Lovell Theorem **18**
11.1 Motivation . . . 18

12 Diagnostic Toolkit Summary **19**

13 Theorem Map: How the Results Connect **20**

14 Conclusion **21**

1 Introduction

Regression is the most widely used tool in quantitative data analysis. It produces point estimates, standard errors, confidence intervals, p-values, and R^2 statistics—all of which look authoritative. In most observational analyses, at least one of these numbers is wrong.

These lecture notes formalise the mathematical foundations behind the eight core failure modes of ordinary least squares (OLS) regression. Each section corresponds to one interactive notebook in the accompanying lab series. The notes are designed to be self-contained: every claim is stated as a formal result and, in the lab, confirmed by Monte Carlo simulation.

Each section follows a consistent pattern: we begin with an everyday story that makes the problem visceral, then state the rigorous mathematics, then work through a tiny numerical example you can verify on paper, and finally describe an interactive visualization that lets you play with the result.

1.1 The NovaMart Scenario

Throughout the course, we work with a single fictional company—**NovaMart**, an online retailer. The variables of interest are:

- Y: monthly revenue (thousands of dollars),
- X_1: advertising spend (thousands of dollars),
- X_2: number of sales staff,
- X_3: customer satisfaction score,
- U: unobserved market demand (hidden from the analyst).

Using a single recurring dataset creates continuity: when a coefficient changes from one notebook to the next, the student remembers the old value.

2 The Data Generating Process

Definition 2.1 (NovaMart DGP)**.** The data are generated according to the following structural equations:

$$U \sim \mathcal{N}(0,1), \tag{1}$$

$$X_1 = \gamma_1 U + \eta_1, \quad \eta_1 \sim \mathcal{N}(0, \sigma_\eta^2), \tag{2}$$

$$X_2 = \gamma_2 U + \eta_2, \quad \eta_2 \sim \mathcal{N}(0, \sigma_\eta^2), \tag{3}$$

$$\varepsilon \sim \mathcal{N}(0,\ h \cdot |X_1|)\,, \tag{4}$$

$$Y = \beta_0 + \beta_1 \log(X_1) + \beta_2 X_2 + \beta_U U + \varepsilon, \tag{5}$$

$$X_3 = \alpha\, Y + \eta_3, \quad \eta_3 \sim \mathcal{N}(0, \sigma_\eta^2). \tag{6}$$

If these structural equations look intimidating, don't worry. Each section ahead will isolate one piece and make it tangible through stories and examples before we prove anything.

Remark 2.1 (Probability limit)*.* Throughout these notes, plim denotes the *probability limit*—the value an estimator converges to as the sample size grows to infinity. Think of it as "what the estimator settles down to with unlimited data."

The default parameter values are:

Parameter	Symbol	Default
Intercept	β_0	50
Ad spend effect	β_1	8
Staff effect	β_2	3
Demand effect	β_U	2
Endogeneity strength	γ_1	20
Staff–demand loading	γ_2	5
Heteroscedasticity	h	0.5
Bad-control strength	α	0.1
Noise scale	σ_η	1

Remark 2.2 (Causal ordering). The generation order is critical: first U, then (X_1, X_2), then Y, then X_3. The variable X_3 is *post-treatment*: it is caused by Y, not a cause of Y. It never appears in the true DGP for Y. Including X_3 as a control variable introduces collider bias (see Section 8).

2.1 Pathologies Embedded in the DGP

The single DGP simultaneously contains all of the following pathologies:

1. **Omitted variable bias:** U drives both X_1 and Y. Omitting U biases $\hat{\beta}_1$ upward.
2. **Heteroscedasticity:** $\mathrm{Var}(\varepsilon \mid X_1) = h \cdot |X_1|$; larger campaigns have more variable returns.
3. **Inflated significance:** with large n, trivially small effects become "significant."
4. **Misspecification:** the true relationship is $\log(X_1)$, not X_1.
5. **Overfitting:** adding polynomial terms inflates training R^2 but destroys test R^2.
6. **Collider bias / bad control:** X_3 is caused by Y; conditioning on it opens a spurious path.

Each notebook isolates and examines one pathology while the others remain present in the background.

3 Why Your Coefficient Is Wrong: Omitted Variable Bias

3.1 Setup

The Sunscreen Murder Mystery

Every summer, ice cream sales go up and so do drowning deaths. If you just look at the data, it seems like ice cream *causes* drownings. But you forgot about the hot weather—when it's hot, people buy more ice cream AND swim more. The heat is doing the real work behind the scenes, but because you ignored it, its effect got smeared onto ice cream, making ice cream look dangerous.

This is exactly what the OVB formula captures: the bias on ice cream's coefficient equals the true effect of temperature on drownings times the correlation between temperature and ice cream sales. Leave out a variable that matters and is correlated with what you included, and its ghost haunts your estimate.

Suppose the true model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X_1, X_2] = 0, \tag{7}$$

but the analyst estimates the *short* regression

$$Y = \tilde{\beta}_0 + \tilde{\beta}_1 X_1 + \tilde{\varepsilon}. \tag{8}$$

Theorem 3.1 (Omitted Variable Bias Formula)**.** *Let $\hat{\beta}_1^{short}$ denote the OLS estimator of $\tilde{\beta}_1$ from the short regression (8), and let β_1 be the true coefficient in (7). Define the* auxiliary slope

$$\delta_1 \;=\; \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_1)}.$$

Then

$$\mathbb{E}[\hat{\beta}_1^{\text{short}}] \;=\; \beta_1 + \beta_2\,\delta_1. \tag{9}$$

The term $\beta_2\delta_1$ is the ***omitted variable bias****.*

Proof. Write the OLS estimator in population form. From the short regression,

$$\text{plim}\ \hat{\beta}_1^{\text{short}} = \frac{\mathrm{Cov}(X_1, Y)}{\mathrm{Var}(X_1)} = \frac{\mathrm{Cov}(X_1,\ \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon)}{\mathrm{Var}(X_1)}.$$

Since $\mathbb{E}[\varepsilon \mid X_1, X_2] = 0$ implies $\mathrm{Cov}(X_1, \varepsilon) = 0$, we obtain

$$\text{plim}\ \hat{\beta}_1^{\text{short}} = \beta_1 + \beta_2 \cdot \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_1)} = \beta_1 + \beta_2\,\delta_1. \qquad \square$$

Studying, Sleeping, and Exam Scores

Five students have data on hours studied (X_1), hours slept (X_2), and exam scores (Y). The true model is $Y = 60 + 5X_1 + 3X_2$ (no noise).

Student	X_1 (studied)	X_2 (slept)	Y (score)
A	1	8	89
B	2	6	88
C	3	7	96
D	4	4	92
E	5	5	100

Full regression (include both X_1 and X_2): recovers $\hat{\beta}_1 = 5$, $\hat{\beta}_2 = 3$.
Short regression (omit sleep): gives $\hat{\beta}_1^{\text{short}} = 2.6$.
Auxiliary regression (X_2 on X_1): slope $\delta_1 = -0.8$ (each extra hour of study costs 0.8 hours of sleep).
Verify the OVB formula:

$$\hat{\beta}_1^{\text{short}} = \beta_1 + \beta_2 \cdot \delta_1 = 5 + 3 \times (-0.8) = 5 - 2.4 = 2.6. \quad \checkmark$$

Omitting sleep shrank the study coefficient from 5 to 2.6—each hour of study costs sleep, and sleep helps scores, so some of study's real effect gets "charged" to the missing sleep variable.

"The Ghost in the Regression" — Interactive OVB Explorer

Try it: A scatterplot of X_1 vs. Y shows two regression lines—the full model (gold) and the short model (red). Toggle a switch to include/exclude X_2 and watch the slope shift in real time.
Key interactions:

- *Correlation slider* (-0.95 to $+0.95$): controls how strongly X_1 and X_2 are related. At zero correlation, both lines coincide—omitting an uncorrelated variable causes no bias.

- *Effect-size slider* (β_2 from -5 to $+10$): when the omitted variable doesn't matter ($\beta_2 = 0$), there's no bias regardless of correlation.

The formula panel updates live, showing how $\beta_2 \cdot \delta_1$ produces the exact gap you see.

Key Insight

The bias has a *sign* and a *direction*. In the NovaMart DGP with default parameters, $\beta_2 = 3$ and $\delta_1 > 0$ (because both X_1 and X_2 are driven by U), so the bias is *positive*: the short regression overestimates the effect of ad spend. The OVB formula predicts $\mathbb{E}[\hat{\beta}_1^{\text{short}}] \approx 2.3 + 2.0 \times 1.2 = 4.7$, which is confirmed by Monte Carlo simulation (mean ≈ 4.698 over 5,000 replications).

3.2 When OVB Cannot Be Signed

The formula (9) generalises to multiple omitted variables. If k variables are omitted, the bias on $\hat{\beta}_1$ is a weighted sum of terms $\beta_j \delta_{1j}$, where each δ_{1j} is the partial regression coefficient of X_j on X_1 after partialling out included regressors. In general, the sign of the total bias is indeterminate without knowledge of all omitted variables.

3.3 The Proxy Variable Approach

The Repair

If the true confounder U is unobservable, one can include a *proxy* $\tilde{U}$ that is correlated with U. If $\tilde{U}$ explains a fraction $q \in [0, 1]$ of the variation in U, the residual bias is approximately $(1 - q)$ times the original bias. At $q = 1$ (perfect proxy), the bias vanishes; at $q = 0$, the full bias remains.

Warning

Including X_3 (customer satisfaction) as a control makes things *worse*, not better. Because X_3 is caused by Y, conditioning on X_3 opens a collider path and introduces additional bias.

4 Why Your Uncertainty Is Wrong: Heteroscedasticity

4.1 The Classical Variance Formula

The Bathroom Scale That Lies About Its Own Reliability

You have a bathroom scale that gives the right weight *on average*—step on it 100 times and the average is your true weight. But when you weigh a kitten, the readings barely wobble. When you weigh a suitcase full of books, the readings scatter all over the place. If you pretend the wobble is the same for everything, you'll think you know the suitcase's weight much more precisely than you actually do. The scale isn't biased—it's centered on the truth—but your confidence about "how close am I?" is completely wrong.

This is what the sandwich variance formula fixes: when the noise differs across observations, the simple formula for uncertainty is wrong, and you need the corrected version to get honest error bars.

Under the standard linear model $Y = X\beta + \varepsilon$ with $\mathbb{E}[\varepsilon\varepsilon^\top \mid X] = \sigma^2 I_n$, the variance of the OLS estimator is

$$\mathrm{Var}(\hat{\beta}_{\mathrm{OLS}} \mid X) = \sigma^2 (X^\top X)^{-1}. \tag{10}$$

Definition 4.1 (Heteroscedasticity)**.** The error term is *heteroscedastic* if $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma_i^2$ with σ_i^2 not constant across i. In the NovaMart DGP, $\sigma_i^2 = h \cdot |X_{1,i}|$: larger ad campaigns have more unpredictable returns.

Theorem 4.1 (Variance Under Heteroscedasticity)**.** *When $\mathbb{E}[\varepsilon\varepsilon^\top \mid X] = \Omega \neq \sigma^2 I_n$, the OLS estimator $\hat{\beta}_{OLS}$ remains* unbiased *but its variance becomes*

$$\mathrm{Var}(\hat{\beta}_{OLS} \mid X) = (X^\top X)^{-1} X^\top \Omega\, X\, (X^\top X)^{-1}. \tag{11}$$

*This is the **sandwich formula**. The classical formula (10) underestimates or overestimates the true variance, depending on the correlation structure between X and the error variances.*

Five Companies, One Funnel

We predict revenue (Y, in \$K) from ad spend (X, in \$K) for five companies:

Company	X	Y	$\hat{Y} = 2.2X$	Residual $\hat{\varepsilon}$	$\hat{\varepsilon}^2$
A	1	3	2.2	+0.8	0.64
B	2	5	4.4	+0.6	0.36
C	3	7	6.6	+0.4	0.16
D	4	3	8.8	−5.8	33.64
E	5	15	11.0	+4.0	16.00

Low-X companies (A, B) have average $\hat{\varepsilon}^2 = 0.50$. High-$X$ companies (D, E) have average $\hat{\varepsilon}^2 = 24.82$—**fifty times larger**. The classical SE of 1.30 treats all five companies as equally informative. But the big spenders are wildly noisier, so the classical confidence interval is a lie.

"The Funnel of False Confidence" — Coverage Simulator

Try it: A scatterplot shows data fanning out (the funnel). Two confidence bands overlay the regression line: classical (constant width, blue) and robust (varying width, green).
Key interaction: Click "Draw 100 new samples." Watch 100 confidence intervals cascade in, coloring green if they contain the truth and red if they miss. With heteroscedasticity, classical CIs achieve only ≈82% coverage while robust CIs hit the promised 95%.
Break it: Set sample size to $n = 20$ and watch the robust SEs *perform worse* than classical—the repair has a minimum sample-size cost.

4.2 Consequences for Inference

If the classical standard errors are used under heteroscedasticity:

- Confidence intervals have incorrect coverage. In the NovaMart simulation, the nominal 95% CI achieves only ≈82% coverage.
- t-statistics and p-values are unreliable.
- The coefficient estimates themselves are *not* biased—only the uncertainty is wrong.

4.3 The Breusch–Pagan Test

Definition 4.2 (Breusch–Pagan Test)**.** Regress the squared OLS residuals $\hat{\varepsilon}_i^2$ on the regressors X. Under H_0: homoscedasticity, the test statistic nR^2 from this auxiliary regression follows χ^2_p asymptotically, where p is the number of regressors (excluding the constant).

4.4 Robust Standard Errors

The Repair

The **White–Huber–Eicker** ("sandwich") standard errors are computed as

$$\widehat{\mathrm{Var}}(\hat{\beta}_{\mathrm{OLS}}) = (X^\top X)^{-1} \left(\sum_{i=1}^{n} \hat{\varepsilon}_i^2 \, x_i x_i^\top \right) (X^\top X)^{-1},$$

where $\hat{\varepsilon}_i$ are the OLS residuals. These standard errors are consistent under heteroscedasticity of unknown form. With robust SEs, the 95% CI recovers ≈95% coverage.

Warning

Robust standard errors perform poorly in small samples. When $n = 20$, the sandwich estimator is more biased than the classical estimator. The crossover point depends on the severity of heteroscedasticity, but as a rough guide, $n \geq 50$ is typically needed for reliable sandwich SEs.

5 Why Your Significance Is Wrong: Effect Size and Power

5.1 The Anatomy of a t-Statistic

The Coin That's Almost Fair

Imagine a coin that lands heads 50.1% of the time instead of exactly 50%. Flip it 20 times and you'd never notice—it looks perfectly fair. Flip it 10 million times and you can prove beyond any doubt that it's rigged. But so what? It's still basically fair—no casino would care, no bet would change. You "detected" a difference so tiny it means nothing in real life, just because you had an ocean of data. Statistical significance told you "it's not exactly 50%," but it never told you "and you should care."
This is exactly what the t-statistic scaling with $\sqrt{n}$ shows: for any nonzero effect, no matter how trivial, collecting enough data will eventually make it "statistically significant"—which is why significance is not the same as importance.

The t-statistic for testing $H_0 : \beta_j = 0$ is

$$t = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)}. \tag{12}$$

Since $\mathrm{SE}(\hat{\beta}_j) = O(1/\sqrt{n})$, for any fixed $\beta_j \neq 0$:

$$|t| \;\to\; \infty \quad \text{as } n \to \infty.$$

Key Insight

With enough data, *every* non-zero effect becomes statistically significant, no matter how small. With $n = 50{,}000$ observations (NovaMart at store-month level), customer satisfaction has $p < 0.001$ despite a coefficient of \$0.003 per satisfaction point—economically meaningless.

5.2 Statistical Power

Definition 5.1 (Power)**.** The **power** of a test is the probability of correctly rejecting H_0 when H_0 is false:

$$\text{Power} = P(|t| > c_\alpha \mid \beta_j \neq 0)\,,$$

where c_α is the critical value at significance level α.

Proposition 5.1 (Power Increases with n and Effect Size)**.** *For a two-sided t-test at level α with true coefficient β_j and standard deviation σ,*

$$\textit{Power} \approx \Phi\left(\frac{|\beta_j|}{\sigma/\sqrt{n}} - z_{\alpha/2}\right),$$

where Φ is the standard normal CDF and $z_{\alpha/2}$ is the upper $\alpha/2$ quantile. Power increases with both $|\beta_j|$ (effect size) and $\sqrt{n}$.

5.3 The Multiple Testing Problem

Running m independent hypothesis tests at level α produces

$$\mathbb{E}[\text{false positives}] = m \cdot \alpha.$$

With $m = 1{,}000{,}000$ tests at $\alpha = 0.05$, we expect 50,000 false positives.

The Repair

The **Bonferroni correction** tests each hypothesis at level α/m, controlling the family-wise error rate at α. For genome-wide association studies ($m \approx 10^6$), the corrected threshold is $\alpha/m = 5 \times 10^{-8}$. Always report effect sizes and confidence intervals alongside p-values.

The Keyboard That Saves 0.1 Seconds

A new keyboard layout saves an average of $\beta = 0.1$ seconds per email ($\sigma = 2$ seconds). Is this "significant"?

Sample size n	SE $= \sigma/\sqrt{n}$	$t = 0.1/\text{SE}$	Significant?
5	0.894	0.11	No
50	0.283	0.35	No
5,000	0.028	3.54	Yes ($p < 0.001$)

The effect never changed—it was always 0.1 seconds. Only the sample size grew. With 5,000 emails, saving one tenth of a second becomes "highly significant" ($p < 0.001$). Statistical significance tells you "the effect is not zero." It says nothing about whether the effect matters.

"The Magnifying Glass That Can't Measure" — Sample Size Simulator

Try it: A slider controls sample size n (50 to 100,000). Three linked panels show the $|t|$-statistic rising, the p-value plummeting, and the confidence interval narrowing—all while the effect size stays fixed at a tiny value.
The aha moment: The dashboard goes from gray "not significant" to screaming red "HIGHLY SIGNIFICANT ($p < 0.001$)" while the actual effect label never changes. Sig-

nificance is a property of n, not of importance.

6 Why Your Model Is Wrong: Specification Error

6.1 The Problem of Functional Form

The Straight Ruler on a Winding Road

You're driving up a mountain on a road that curves back and forth. A friend asks you to describe the road, but you only have a straight ruler. You lay it on the map across the section you've driven so far, and it gives a decent summary—"on average, the road goes uphill at about this angle." But then you extend that straight line past where you've actually been, and it says you should be 500 feet in the air by now, flying off a cliff. The ruler was the *best* straight summary of the curvy road you had data for, but using it to predict beyond that range was a disaster.
This is exactly what the linear projection under misspecification gives you: OLS finds the best-fitting straight line through curved reality, which can look reasonable in-sample but diverges catastrophically when you extrapolate.

Suppose the true relationship is $Y = f(X) + \varepsilon$ for some nonlinear function f, but we fit the linear model $Y = X\beta + \varepsilon$.

Theorem 6.1 (Linear Projection Under Misspecification)**.** *If $Y = f(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X] = 0$, and we estimate $Y = X\beta + u$ by OLS, then the OLS estimator converges to*

$$\beta^* = \arg\min_{\beta} \mathbb{E}\Big[(Y - X\beta)^2\Big],$$

which is the best linear approximation to $f(X)$ in the mean-squared-error sense. This β^* equals the true β only if f is linear.

In the NovaMart DGP, $f(X_1) = \beta_1 \log(X_1)$. A linear model in X_1 provides a reasonable approximation within the observed data range but diverges catastrophically when extrapolating.

6.2 The Extrapolation Danger

If the true relationship is $Y = 8\log(X_1) + \cdots$ and we fit $Y = \tilde{\beta}_1 X_1 + \cdots$:

- At $X_1 = 100$ (within sample): moderate overestimation.
- At $X_1 = 2{,}000$ (extrapolation): linear prediction overshoots by a factor of $\approx$3.
- The confidence band widens but does *not* contain the truth.

The Repair

Fit the correctly specified model by transforming the predictor:

$$Y = \beta_0 + \beta_1 \log(X_1) + \beta_2 X_2 + \varepsilon.$$

Extrapolation accuracy improves dramatically. However, even the log model is a local approximation—at extreme extrapolation, all parametric models eventually fail.

Square Plots and Fence Lengths

The true relationship between the area of a square plot (X, in m^2) and its side length ($Y = \sqrt{X}$, in m) is nonlinear. We fit a straight line to five plots:

Plot	X (area)	True $Y = \sqrt{X}$	Fitted $\hat{Y} = 1.24 + 0.16X$
A	1	1.00	1.40
B	4	2.00	1.88
C	9	3.00	2.68
D	16	4.00	3.80
E	25	5.00	5.25

Maximum in-sample error: just 0.40. Looks fine! Now extrapolate:

- At $X = 100$: line predicts $\hat{Y} = 17.3$. Truth: $\sqrt{100} = 10$. Off by 73%.
- At $X = 10{,}000$: line predicts $\hat{Y} = 1{,}606$. Truth: $\sqrt{10{,}000} = 100$. Off by a factor of 16.

The best linear approximation is only "best" within the range where you asked the question.

"The Comfort Zone" — Extrapolation Explorer

Try it: Drag a prediction point along the x-axis. Within the data range, the background is calm blue and the error bar is tiny. Drag past the data boundary: the background turns progressively red, the error bar explodes, and the warning escalates from "Mild extrapolation" to "CATASTROPHIC: prediction is fiction."
Toggle: Show the correctly specified log model alongside the line. The log model tracks truth even far out—the contrast is stark.

7 Why Your R^2 Is Lying: Overfitting

7.1 The Bias–Variance Decomposition

The Three Dartboard Players

Three friends play darts. The first always throws at the same wrong spot—consistently two inches left of the bullseye. Every throw lands in the same cluster, just in the wrong place. That's high bias, low variance. The second friend aims perfectly and averages a bullseye, but her throws scatter wildly all over the board. That's low bias, high variance. The third friend accepts that the dartboard sits on a wobbly wall, so even a perfect throw gets jostled randomly. That wobble is irreducible noise—nobody can beat it. The best player is the one who aims at the right spot AND throws consistently, finding the sweet spot between the first two friends' mistakes.
This is exactly what the bias-variance decomposition says: total prediction error equals systematic mis-aim (bias squared) plus scatter in your throws (variance) plus the wobble you can never control (irreducible noise).

Theorem 7.1 (Bias–Variance Decomposition)**.** *For any estimator $\hat{f}$ of the regression function f, the expected prediction error at a point x_0 decomposes as*

$$\mathbb{E}\Big[(Y - \hat{f}(x_0))^2\Big] = \underbrace{\Big(f(x_0) - \mathbb{E}[\hat{f}(x_0)]\Big)^2}_{\textit{Bias}^2} + \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\textit{Variance}} + \underbrace{\sigma_\varepsilon^2}_{\textit{Irreducible noise}} . \tag{13}$$

Proof. Let $\mu = \mathbb{E}[\hat{f}(x_0)]$. Then

$$\begin{aligned}\mathbb{E}[(Y - \hat{f})^2] &= \mathbb{E}[(f + \varepsilon - \hat{f})^2] \\ &= \mathbb{E}[(f - \hat{f})^2] + \mathbb{E}[\varepsilon^2] + 2\mathbb{E}[\varepsilon(f - \hat{f})] \\ &= \mathbb{E}[(f - \hat{f})^2] + \sigma_\varepsilon^2,\end{aligned}$$

where the cross-term vanishes because ε is independent of $\hat{f}$ (trained on separate data) and has mean zero. Now expand:

$$\mathbb{E}[(f - \hat{f})^2] = (f - \mu)^2 + \mathbb{E}[(\hat{f} - \mu)^2] = \mathrm{Bias}^2(\hat{f}) + \mathrm{Var}(\hat{f}).$$ □

7.2 Overfitting in Practice

Adding polynomial terms to a regression always increases training R^2 (since R^2 is monotonically non-decreasing in the number of regressors). In the NovaMart demo:

Degree	**Training R^2**	**Test R^2**
1	0.71	0.69
3	0.82	0.78
7	0.91	0.63
15	0.97	< 0

A degree-15 polynomial achieves $R^2 = 0.97$ on training data but predicts *worse than the sample mean* on held-out data.

7.3 Cross-Validation

Definition 7.1 (k-Fold Cross-Validation)**.** Partition the data into k equally sized folds. For each fold j, train the model on all data except fold j and compute the prediction error on fold j. The CV estimate of prediction error is

$$\mathrm{CV}(k) = \frac{1}{k}\sum_{j=1}^{k} \mathrm{MSE}_j\,.$$

The Repair

Using cross-validation, the log-linear model (which matches the true DGP) achieves the lowest CV error, outperforming all polynomial models. The right model family matters more than the degree of flexibility.

Warning

Cross-validation itself can overfit if the number of candidate model specifications searched is large. The best CV R^2 inflates with the breadth of the search.

Three Models, Three Training Sets

True relationship: $Y = 2X$. We predict at $X = 6$ (true $Y = 12$) using three training sets of (X, Y) pairs and three models:

	Model A (constant)	**Model B** (line)	**Model C** (degree-4)
Prediction (Set 1)	7	10.8	8.0
Prediction (Set 2)	7	13.2	16.0
Prediction (Set 3)	7	12.2	−13.0
Average prediction	7.0	12.1	3.7
Bias^2	25.0	≈ 0	69.4
Variance	0.0	1.0	149.6
Total MSE	**25.0**	**1.0**	**219.0**

Model C has zero training error on every dataset, yet its predictions swing from +16 to −13. The line (Model B) wins decisively. Perfect in-sample fit amplifies noise into wild out-of-sample predictions.

"The U-Shaped Trap" — Bias-Variance Explorer

Try it: A slider controls polynomial degree (1 to 15). The left panel shows a stacked area chart of Bias^2 + Variance + Noise forming the classic U-shaped MSE curve. The right panel overlays 20 fitted curves from different training sets.
The aha moment: At degree 1, twenty nearly identical lines all miss the true curve (stable but wrong). At degree 15, twenty polynomials pass through their training points perfectly but look completely different from each other—wild oscillations everywhere (unstable and absurd). Press "Animate" and watch the sweet spot at degree 3–4.

8 Why Your Causal Claim Is Wrong: Endogeneity and Bad Controls

8.1 Endogeneity

Why Every Movie Star's Spouse Seems Boring

Among famous actors, their spouses seem to be unusually un-funny or un-attractive. Does fame make you marry dull people? No—you're only looking at people who are already famous. People become famous either by being extremely talented or by being extremely good-looking (or both). Once you filter for "is famous," the talented-but-plain people and the gorgeous-but-untalented people get mixed together, creating a fake tradeoff that doesn't exist in the general population. By restricting your attention to a group defined by the outcome of the very thing you're studying, you manufactured a relationship from thin air.
This is exactly what collider bias describes: when you control for a variable that is *caused by* the things you're studying, you create a spurious association that didn't exist before—more controls made the answer worse, not better.

Definition 8.1 (Endogeneity)**.** A regressor X_j is *endogenous* if $\mathrm{Cov}(X_j, \varepsilon) \neq 0$. This can arise from omitted variables, measurement error, or simultaneous causation (feedback loops).

In the NovaMart DGP, revenue and ad spend are mutually related through demand U. OLS captures the correlation but cannot distinguish causal direction.

8.2 Collider Bias from Bad Controls

The causal graph for NovaMart is:

$$U \longrightarrow X_1 \longrightarrow Y \longrightarrow X_3, \qquad U \longrightarrow Y.$$

Including X_3 (satisfaction) as a regressor conditions on a descendant of Y, which opens a spurious association path through Y.

Proposition 8.1 (Collider Bias)**.** *If $X_3 = \alpha Y + \eta_3$ (i.e., X_3 is a post-treatment variable), then the regression of Y on (X_1, X_3) yields a biased estimate of the causal effect of X_1 on Y, even if X_1 is otherwise exogenous. The bias arises because conditioning on a descendant of Y partially conditions on Y itself.*

Warning

The advice "control for everything available" is wrong. Controlling for post-treatment variables or colliders *introduces* bias rather than removing it. Variable selection for causal inference requires understanding the causal structure, not just statistical association.

Talent, Luck, and the Fame Filter

People have Talent (0 or 1) and Luck (0 or 1), independently assigned. A person is Famous if they have Talent OR Luck (or both).

Type	Talent	Luck	Famous?
I	0	0	No
II	0	1	Yes
III	1	0	Yes
IV	1	1	Yes

Full population: $P(\text{Talent} = 1) = 1/2$, $P(\text{Luck} = 1) = 1/2$, and $P(\text{both}) = 1/4 = 1/2 \times 1/2$. Talent and Luck are **independent**. Correlation $= 0$.
Among the Famous only (drop Type I): If someone is talented (Types III, IV), $P(\text{Lucky}) = 1/2$. If not talented (Type II only), $P(\text{Lucky}) = 1$. Suddenly, knowing a famous person is talented makes them *less* likely to be lucky!
The observation "famous people are either talented OR lucky but rarely both" is not evidence that the universe is fair—it is a mathematical artifact of selecting on a collider.

"The Paradox of Conditioning" — Collider Explorer

Try it: A scatterplot of Talent vs. Luck for 1,000 people. All dots are gray; no correlation visible. Toggle "Condition on Fame" and watch: most dots fade, the highlighted subset reveals a clear negative slope. A DAG panel alongside animates: a box appears around the Fame node, and a dashed red "spurious association" arc draws between Talent and Luck.
Slider: Fame threshold—from 10th to 90th percentile. At low thresholds (most people famous), the spurious correlation is weak. At high thresholds (top 5%), it becomes extreme.

8.3 Instrumental Variables

Definition 8.2 (Instrument)**.** A variable Z is a valid *instrument* for X in the equation $Y = X\beta + \varepsilon$ if:

1. **Relevance:** $\mathrm{Cov}(Z, X) \neq 0$.
2. **Exclusion:** $\mathrm{Cov}(Z, \varepsilon) = 0$ (the instrument affects Y only through X).

Theorem 8.1 (IV Consistency)**.** *If Z is a valid instrument, the IV estimator*

$$\hat{\beta}_{IV} = (Z^\top X)^{-1} Z^\top Y \tag{14}$$

is consistent for β:

$$\operatorname{plim} \hat{\beta}_{IV} = \operatorname{plim} \frac{Z^\top Y/n}{Z^\top X/n} = \frac{\mathrm{Cov}(Z, Y)}{\mathrm{Cov}(Z, X)} = \beta.$$

In the NovaMart context, if ad prices varied randomly across regions (a natural experiment), the regional price variation serves as an instrument for ad spend.

Warning

Weak instruments ($\mathrm{Cov}(Z, X) \approx 0$) cause the IV estimator to be highly variable and unreliable. When the first-stage F-statistic is below 10, IV confidence intervals are extremely wide and the estimator may be more biased than OLS.

Education, Wages, and Distance to College

We want the causal effect of education (X) on wages (Y, in \$K). Problem: ability (U) boosts both. We use distance to college (Z) as an instrument.

Person	Z (distance)	U (ability)	$X = 2 + Z + U$	$Y = 10 + 3X + U$
A	1	1	4	23
B	2	-2	2	14
C	3	0	5	25
D	4	2	8	36
E	5	-1	6	27

True causal effect: $\beta = 3$. Verify: $\mathrm{Cov}(Z, U) = 0$ by construction.
OLS (biased): slope $= \mathrm{Cov}(X, Y)/\mathrm{Var}(X) = 14/4 = 3.5$. Ability inflates the estimate by $+0.5$.
IV: $\hat{\beta}_{\mathrm{IV}} = \mathrm{Cov}(Z, Y)/\mathrm{Cov}(Z, X) = 6/2 = 3.0$. Exactly the true β!
The instrument strips out ability contamination by using only the variation in education that comes from geography, not brains.

"The Backdoor and the Detour" — IV Strength Simulator

Try it: Three linked panels show the first stage (Z vs X), reduced form (Z vs Y), and structural (X vs Y) relationships. The structural panel overlays OLS (blue, biased), truth (gold), and IV (green) lines.
Key slider: Instrument strength (0.01 to 0.95). With a strong instrument, IV hugs the truth. Drag toward zero: the first-stage scatter becomes noise, the F-statistic drops below 10 (turns red), and the IV sampling distribution *explodes*. A weak instrument is worse than no instrument at all.

9 The Real World: When All Pathologies Strike at Once

9.1 The Lalonde Dataset

The previous sections studied one pathology at a time in a clean simulated world. Real data is not so polite—every pathology arrives simultaneously, and none announces itself.

Notebook 7 transitions from simulated data to the Lalonde (1986) dataset, a benchmark in causal inference. The question: does a job training programme increase earnings?

- **Observational estimate (OLS):** negative coefficient—training appears to *reduce* earnings.
- **Experimental estimate (RCT):** positive and large (≈$1,800).
- The two estimates do not even have the same sign.

All standard diagnostics pass: VIFs are low, Breusch–Pagan is non-significant, robust SEs barely change. The problem is not in the model—it is in the *data*. The observational and experimental groups differ systematically on unobservables.

9.2 Sensitivity Analysis: Cinelli–Hazlett Framework

Definition 9.1 (Robustness Value)**.** The **robustness value** (RV) is the minimum strength of association (measured in partial R^2) that an unobserved confounder would need to have with both the treatment and the outcome to explain away the observed result.

Formally, define

$$\mathrm{RV}_{q,\alpha} = \min\{r : \text{adjusted estimate crosses } q \cdot \hat{\beta} \text{ at level } \alpha\}.$$

A small RV indicates fragility: a weak confounder suffices to nullify the result. A large RV indicates robustness.

Key Insight

The Lalonde observational estimate has a very small robustness value: the result is fragile. A confounder with even modest partial R^2 with both treatment and outcome would explain away the finding. This is why the observational and experimental estimates disagree.

10 The Redemption: Regression Discontinuity Design

10.1 When Design Earns Trust

Regression works brilliantly when the world provides a natural experiment. The **regression discontinuity design** (RDD) exploits a sharp threshold to identify causal effects.

The Kid Who Scored 69 vs. The Kid Who Scored 70

Your school says anyone who scores 70 or above on the final exam gets into the honours class next year. The kid who scored 70 and the kid who scored 69 are basically the same student—maybe one got lucky on one question, maybe the other had a headache. They studied the same amount, have the same ability, live in the same neighbourhoods. But one gets the honours class and the other doesn't. If you come back a year later and the 70-kid is doing noticeably better, that gap can't be explained by the kids being different—they were virtually identical. The jump in outcomes right at the cutoff *is* the real effect of the honours class itself.

This is exactly what the regression discontinuity design measures: the difference in outcomes just above versus just below the cutoff, where people are essentially identical, gives you the causal effect without a randomised experiment.

Definition 10.1 (Sharp RDD)**.** Suppose treatment is assigned deterministically by a *running variable* R at a cutoff c:

$$D_i = \mathbf{1}(R_i \geq c).$$

Under continuity of potential outcomes at c, the **local average treatment effect** at the cutoff is

$$\tau_{\text{RDD}} = \lim_{r \downarrow c} \mathbb{E}[Y \mid R = r] - \lim_{r \uparrow c} \mathbb{E}[Y \mid R = r]. \tag{15}$$

Theorem 10.1 (RDD Identification)**.** *If $\mathbb{E}[Y_0 \mid R = r]$ and $\mathbb{E}[Y_1 \mid R = r]$ are continuous at $r = c$, then*

$$\tau_{RDD} = \mathbb{E}[Y_1 - Y_0 \mid R = c].$$

The design eliminates confounding locally *at the cutoff because units just above and just below the threshold are nearly identical in all characteristics.*

10.2 Example: Scholarship Cutoff

Students with test scores just above a threshold receive scholarships; those just below do not. Near the cutoff, the two groups are essentially identical. The jump in outcomes at the threshold identifies the causal effect of the scholarship.

10.3 Bandwidth Choice

The RDD estimator faces a bias–variance tradeoff controlled by the *bandwidth* h:

- **Narrow bandwidth:** low bias (observations are close to the cutoff), high variance (few observations).
- **Wide bandwidth:** low variance (more observations), high bias (observations far from the cutoff may not be comparable).

Optimal bandwidth selection (e.g., Imbens–Kalyanaraman or Calonico–Cattaneo–Titiunik) balances these two forces.

The Repair

The RDD does not require controlling for confounders. The design itself ensures comparability near the cutoff. When the autopsy report is run on the RDD estimate, all indicators are green: the regression is trustworthy because the *design* is trustworthy.

Scholarship Scores and Outcomes

Students who score ≥ 50 on an entrance exam receive a scholarship. Data generated from $Y = 60 + 0.5R + 5D$ (true treatment effect $\tau = 5$):

Student	R (score)	Scholarship?	Y (outcome)
1	43	No	81.5
2	45	No	82.5
3	47	No	83.5
4	49	No	84.5
5	51	Yes	90.5
6	53	Yes	91.5
7	55	Yes	92.5
8	57	Yes	93.5

Naïve above-vs-below comparison: $92.0 - 83.0 = 9.0$. But this includes the baseline trend (higher-scoring students do better anyway).

Fit lines on each side and extrapolate to the cutoff ($R = 50$): below gives $\hat{Y}(50) = 85.0$, above gives $\hat{Y}(50) = 90.0$. The RD estimate: $\hat{\tau} = 90.0 - 85.0 = 5.0$—exactly the true treatment effect.

"The Cliff" — Bandwidth Explorer

Try it: A scatterplot of Y vs. test score R with a vertical cutoff line. Two local regression lines meet at the cutoff; the gap is the treatment effect. A bandwidth slider (2 to 50) controls how many points are used: narrow bandwidth shows a clean jump with wide CI, wide bandwidth smooths the jump away.
Break it: Drag the cutoff away from the true value (50) and watch the "treatment effect" vanish—a built-in falsification test.

11 What "Controlling For" Actually Means: The Frisch–Waugh–Lovell Theorem

11.1 Motivation

When we say "the coefficient on X_1, controlling for X_2," we do *not* mean "holding X_2 constant." We mean: the coefficient from regressing the part of Y unexplained by X_2 on the part of X_1 unexplained by X_2.

Is It the Coaching or Just the New Shoes?

Your friend claims her new running coach made her faster, but she also bought expensive new running shoes at the same time. How do you figure out what the coach actually contributed? Step one: figure out how much of her speed improvement the shoes alone would predict, and subtract that out—what's left is the part of her speed the shoes *can't* explain. Step two: figure out how much of the coaching hours the shoes can predict (maybe the coach recommended the shoes, so they're connected), and subtract that out too—what's left is the coaching variation that has nothing to do with shoes. Now compare those two leftovers: the unexplained speed against the unexplained coaching. *That* remaining relationship is the coach's true contribution, cleaned of the shoe effect.
This is exactly what the Frisch–Waugh–Lovell theorem proves: "controlling for" shoes means removing shoes' influence from *both* the outcome (speed) and the variable of interest (coaching), then running the regression on the residuals—the double-residualisation gives you the same coefficient you'd get in the full multiple regression.

Theorem 11.1 (Frisch–Waugh–Lovell)**.** *Consider the regression $Y = X_1\beta_1 + X_2\beta_2 + \varepsilon$. Let $M_2 = I - X_2(X_2^\top X_2)^{-1}X_2^\top$ be the residual-maker matrix that projects orthogonally to the column space of X_2. Then*

$$\hat{\beta}_1 = (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y. \tag{16}$$

That is, $\hat{\beta}_1$ from the full regression is identical *(to numerical precision) to the coefficient obtained by:*

1. *Regressing Y on X_2 and taking residuals: $\tilde{Y} = M_2 Y$.*
2. *Regressing X_1 on X_2 and taking residuals: $\tilde{X}_1 = M_2 X_1$.*
3. *Regressing $\tilde{Y}$ on $\tilde{X}_1$ (no intercept).*

Proof. The OLS normal equations for the partitioned model $Y = X_1\beta_1 + X_2\beta_2 + \varepsilon$ give

$$\begin{pmatrix} X_1^\top X_1 & X_1^\top X_2 \\ X_2^\top X_1 & X_2^\top X_2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} X_1^\top Y \\ X_2^\top Y \end{pmatrix}.$$

From the second block, $\hat{\beta}_2 = (X_2^\top X_2)^{-1} X_2^\top (Y - X_1\hat{\beta}_1)$. Substituting into the first block and simplifying:

$$\begin{aligned} X_1^\top X_1 \hat{\beta}_1 + X_1^\top X_2 (X_2^\top X_2)^{-1} X_2^\top (Y - X_1\hat{\beta}_1) &= X_1^\top Y \\ X_1^\top [I - X_2(X_2^\top X_2)^{-1} X_2^\top] X_1 \hat{\beta}_1 &= X_1^\top [I - X_2(X_2^\top X_2)^{-1} X_2^\top] Y \\ X_1^\top M_2 X_1 \hat{\beta}_1 &= X_1^\top M_2 Y. \end{aligned}$$

Hence $\hat{\beta}_1 = (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y$. □

Key Insight

"Controlling for" means residualising—stripping out the linear relationship with the control variables from *both* sides of the equation. The partial regression plot (also called an added-variable plot) visualises this: it plots $\tilde{Y}$ against $\tilde{X}_1$. The slope of this scatterplot *is* $\hat{\beta}_1$ from the full multiple regression.

Exercise, Sleep, and Mood

We want the effect of exercise (X_1, hours) on mood (Y, score 1–20), controlling for sleep (X_2, hours).

Person	X_1 (exercise)	X_2 (sleep)	Y (mood)
A	1	2	3
B	2	4	7
C	3	6	10
D	5	8	18
E	4	10	12

Step 1: Regress Y on X_2: slope $= 1.45$. Residuals $\tilde{Y} = (-1.2,\ -0.1,\ 0.0,\ +5.1,\ -3.8)$.
Step 2: Regress X_1 on X_2: slope $= 0.45$. Residuals $\tilde{X}_1 = (-0.2,\ -0.1,\ 0.0,\ +1.1,\ -0.8)$.
Step 3: Regress $\tilde{Y}$ on $\tilde{X}_1$: slope $= 8.9/1.9 = \mathbf{4.684}$.
Full multiple regression (Y on X_1 and X_2): coefficient on $X_1 = \mathbf{4.684}$.
They match—not approximately, but to every decimal place. "Controlling for sleep" literally means: remove everything sleep can explain from *both* mood and exercise, then see what's left.

"The Double Residual" — FWL Animator

Try it: Three panels. Panel 1: a 3D scatter of (X_1, X_2, Y) with the regression plane. Panel 2: two mini-scatters showing the residualisation (residual bars appear as regressions are fit). Panel 3: the partial regression plot—residuals vs. residuals.
The aha moment: Click "Animate." Watch regression lines fit in Panel 2, residual bars appear, then the residual dots "fly" into Panel 3 and reassemble into a new scatter. The slope reads 4.684. Check the multiple regression output: $\beta_1 = 4.684$. Exact match.

12 Diagnostic Toolkit Summary

Think of these diagnostics as the instrument panel in a cockpit. Each gauge checks one specific condition—but all gauges reading green does not mean the flight plan is correct.

The following diagnostics are used across all notebooks:

Diagnostic	Tests For	Interpretation
Residuals vs. Fitted	Nonlinearity, heteroscedasticity	Residuals should show no pattern; funnel shape indicates heteroscedasticity
Normal Q–Q plot	Non-normality of residuals	Points should lie on the 45° line
Scale–Location	Heteroscedasticity	$\sqrt{\lvert\text{standardised residuals}\rvert}$ should be flat against fitted values
Residuals vs. Leverage	Influential observations	Points with high leverage *and* large residuals (Cook's $d > 1$) are influential
VIF	Multicollinearity	$\text{VIF}_j > 10$ suggests problematic collinearity
Breusch–Pagan	Heteroscedasticity	H_0: homoscedasticity; reject if $p < \alpha$
Durbin–Watson	Autocorrelation	Statistic near 2 indicates no autocorrelation
Jarque–Bera	Non-normality	H_0: residuals are normally distributed

Warning

Passing all diagnostics does not guarantee a valid causal estimate. In the Lalonde example (Notebook 7), every diagnostic passes, yet the OLS estimate has the wrong sign. Diagnostics check the model's internal consistency, not whether the research design is sound.

13 Theorem Map: How the Results Connect

Before the final summary, let us step back and see how all nine results connect. Each theorem in this course relaxes an assumption that the previous one relied on, forming a chain of progressively more honest conclusions.

The theorems in this course form a logical chain where each generalises the previous:

NB	Theorem	Generalisation Relationship
1	Omitted Variable Bias	Foundation: bias from omitting a relevant variable
2	Sandwich Variance	Relaxes $\text{Var}(\varepsilon) = \sigma^2 I$; OVB is a special case of endogeneity
3	Power Analysis	$t = \hat{\beta}/\,\text{SE}(\hat{\beta})$; builds on correct SEs from NB 2
4	Linear Projection	Best linear approx. to $f(X)$; relaxes f linear
5	Bias–Variance Tradeoff	$\text{MSE} = \text{Bias}^2 + \text{Var} + \sigma^2$; generalises specification error
6	IV Consistency	If $\mathbb{E}[\varepsilon X] \neq 0$ but $\mathbb{E}[\varepsilon Z] = 0$; generalises OVB repair
7	Sensitivity Analysis	Robustness value quantifies fragility of any estimate
8	RDD Identification	Continuity at cutoff $\Rightarrow$ causal identification
1.5	Frisch–Waugh–Lovell	What "controlling for" means algebraically

The meta-lesson: each theorem relaxes an assumption that justified the previous conclusion. Classical variance is a special case of the sandwich estimator. The linear model is a special case of all models. OVB is a special case of endogeneity.

14 Conclusion

These notes have formalised the eight core failure modes of OLS regression demonstrated in the Regression Autopsy lab series:

1. Omitted variable bias shifts your coefficient.
2. Heteroscedasticity corrupts your standard errors.
3. Large samples make trivial effects "significant."
4. Misspecification makes extrapolation fiction.
5. Overfitting makes memorisation look like learning.
6. Bad controls and endogeneity destroy causal claims.
7. Real data confronts you with all failures simultaneously.
8. Sound research design (RDD) earns regression its trust back.

The unifying lesson is not that regression should be abandoned, but that its outputs must be accompanied by an honest assessment of the assumptions under which they are valid. The diagnostics tell you whether the *model* is internally consistent; only the *research design* can tell you whether the estimate is credible.

Key Insight

Every section in these notes followed the same arc: a toy story made you feel the problem, the theorem made it precise, a numerical example made it concrete, and a visualisation let you play with it. If you remember only one thing, remember this: regression is not a magic box that turns data into truth. It is a tool with assumptions, and every number it produces is only as honest as those assumptions. Your job is not to believe the output—it is to interrogate it.

This doesn't mean you should stop using regression.
It means you're finally ready to start.